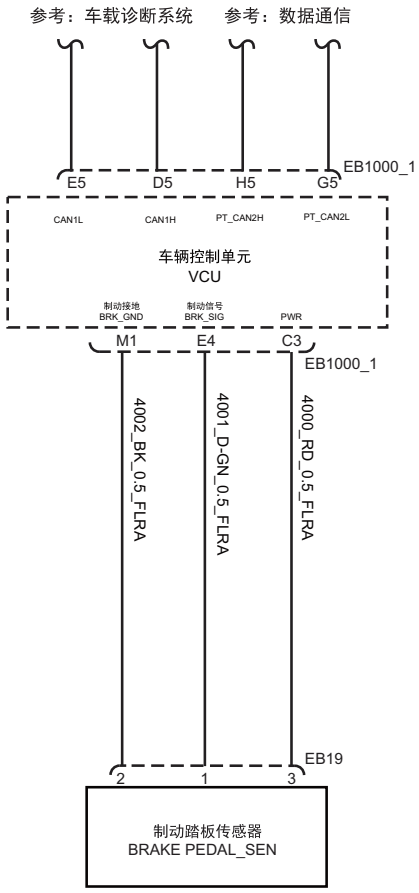
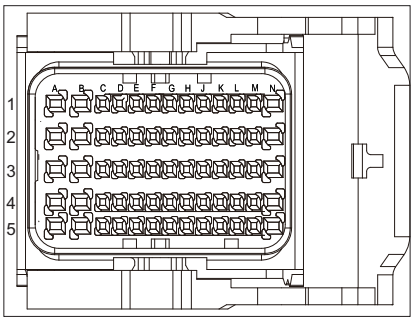
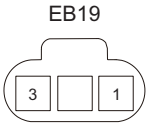
B1203-11、 B1203-12、 B1203-13、 B1203-1E

Fault diagnosis code
B1203-11：Brake pedal circuit short to ground
B1203-12：Brake pedal circuit short to battery
B1203-13：Brake pedal circuit open
B1203-1E：Brake pedal circuit resistance out of range
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BCM fault • Brake pedal sensor fault. • Insurance fuse fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1203-11
Brake pedal circuit short to ground
B1203-12
Brake pedal circuit short to battery
B1203-13
Brake pedal circuit open
B1203-1E
The input resistance of brake pedal circuit is detected to be out of range for more than 1s.
To set the effect of a fault code condition
1 Brake pedal circuit fails to effect 2 Brake pedal circuit function fails to effect
Description of circuit diagram
The VCU module monitors the brake pedal sensor via low level reference circuit and signal circuit. The BCM module receives and monitors the signals of brake pedal sensor via network signals.

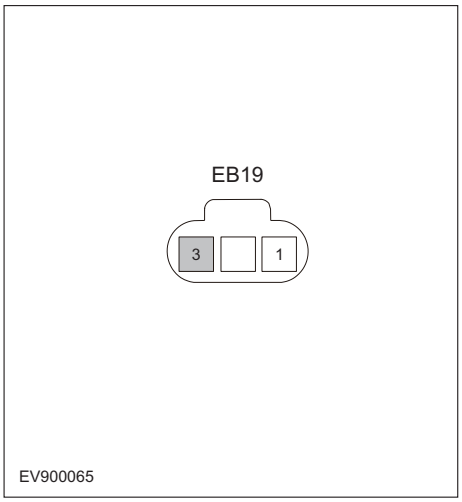
Circuit diagram

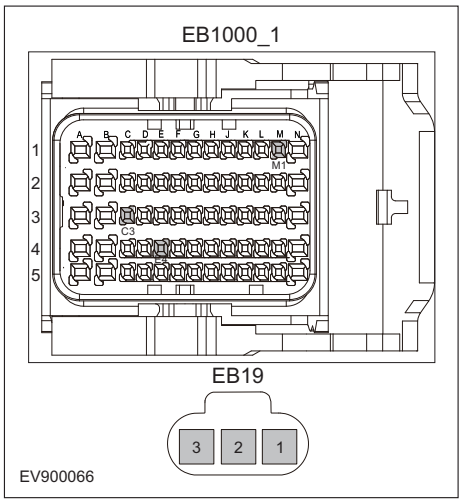


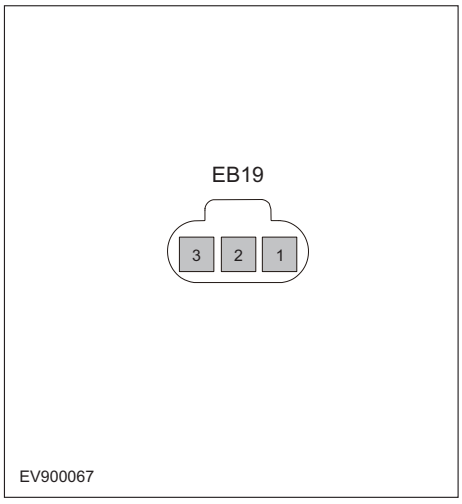


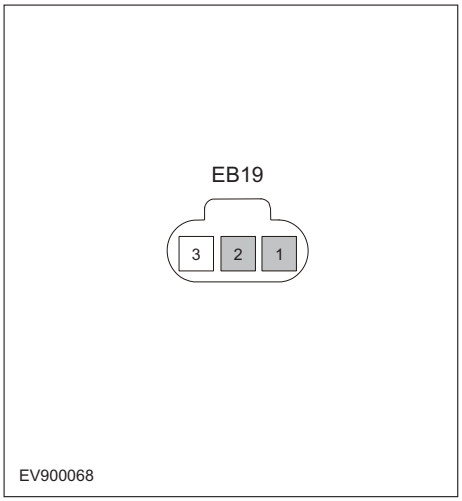


EV900064

<table>
<tr><td colspan="3">Troubleshooting steps</td></tr>
<tr><td>Step 1</td><td colspan="2">General inspection.</td></tr>
<tr><td colspan="2"></td><td>1 Check the fuse for damage.
2 Check battery capacity.
3 Check the harness plug of air conditioning control module for damage, poor contact, aging and looseness.
4 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO ➞ Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
<tr><td>Step 2</td><td colspan="2">Check fault codes.</td></tr>
<tr><td colspan="2"></td><td>1 Connect the fault diagnosis instrument to the diagnosis interface.
2 Turn the ignition switch to ON position.
3 Check the body control module with a diagnostic apparatus.
4 Is there any other fault code except for B1203-11、B1203-12、 B1203-13、 B1203-1E?</td></tr>
<tr><td colspan="2"></td><td>YES ➞ Refer to:DTC Summary list(BCM).</td></tr>
<tr><td colspan="3">NO ↓</td></tr>
<tr><td>Step 3</td><td colspan="2">Check the power supply circuit of brake pedal sensor circuit.</td></tr>
<tr><td colspan="2"> 
</td><td>1 Turn the ignition switch to OFF.
2 Disconnect the harness plug EB19 of brake pedal sensor..
3 Turn the ignition switch to ON.
4 Measure the voltage between the plug terminal of brake pedal sensor and ground.
Measuring circuit: voltage between terminal 3 on plug EB19 and ground.
Standard value:4.8~5.2V
5 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO ➞ Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
</table>

Step 4	Check the circuit between brake pedal sensor and VCU (open).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1000_1 of VCU. 3 Disconnect the harness plug EB19 of brake pedal sensor. 4 Measure the resistance between the harness terminal of brake pedal sensor and the VCU. Measuring circuit: resistance between terminal 1 on plug EB19 and terminal E4 on plug EB1000_1. Measuring circuit: resistance between terminal 2 on plug EB19 and terminal M1 on plug EB1000_1. Measuring circuit: resistance between terminal 3 on plug EB19 and terminal C3 on plug EB1000_1. **Standard value: ≈ 0Ω** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check the power supply circuit of brake pedal sensor circuit.
	1 Turn the ignition switch to OFF. 2 Connect the harness plug EB1000_1 of VCU. 3 Measure the resistance between the harness plug of brake pedal sensor and ground. Measuring circuit: resistance between terminal 1 on plug EB19 and ground. Measuring circuit: resistance between terminal 2 on plug EB19 and ground. Measuring circuit: resistance between terminal 3 on plug EB19 and ground. **Standard value: ∞** 4 Measure the voltage between the harness plug of brake pedal sensor and ground. Measuring circuit: voltage between terminal 1 on plug EB19 and ground. Measuring circuit: voltage between terminal 2 on plug EB19 and ground. Measuring circuit: voltage between terminal 3 on plug EB19 and ground. **Standard value: ≈ 0V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the brake pedal sensor (components).
	1 Turn the ignition switch to OFF. 2 Measure the resistance of the terminal of brake pedal sensor directly （sensor end). Measuring circuit: resistance between terminal 1 and terminal 2 on plug EB19. **Standard value:** 3 Measure the resistance between the brake pedal sensor terminal and the housing. Measuring circuit: resistance between terminal 1 on plug EB19 and the housing. Measuring circuit: resistance between terminal 2 on plug EB19 and the housing. **Standard value: ∞** 4 Check whether the result is normal or not?
	NO → Replace brake pedal sensor.
YES ↓	
Step 7	Test whether BCM is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BCM.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	